Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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This document provides a basic paper template and submission guidelines. If using this (optional) template, your short abstract should go here.

1 Section example

This is a section with an itemized list.

- Item 1
- Item 2

1.1 Subsection example

This is a subsection.

1.1.1 Subsubsectione example

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2 Submitting Final Camera-Ready Copy

Since our workshop is interdisciplinary, we do not require any specific final template for camera-ready. For example, if you work primarily in law, please feel free to submit a PDF version of a standard law-reviewe formatted paper. If your paper was a Fast Track paper that was already accepted for publication elsewhere, there is no need to re-format the paper using this template (unless you want to). You are free to submit a PDF of the paper that was already accepted for publication, in the format of that publication venue. We will label the paper as a Fast Track paper on the website.

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2.1 Camera-Ready Author Information

Affiliations should be numbered in the order of appearance. A single footnote block of text should be used to list all the affiliations. (Academic affiliations should list Department, University, City, State/Region, Country. Similarly for industrial affiliations.)

Each distinct affiliations should be listed once. If an author has multiple affiliations, multiple superscripts should be placed after the name, separated by thin spaces. If the authors would like to highlight equal contribution by multiple first authors, those authors should have an asterisk placed after their name in superscript, and the term "*Equal contribution" should be placed in the footnote block ahead of the list of affiliations. A list of corresponding authors and their emails (in the format Full Name <email@domain.com>) can follow the list of affiliations. Ideally only one or two names should be listed.

A sample file with author names is included

in the ICML2024 style file package. Turn on the `[accepted]` option to the stylefile to see the names rendered. All of the guidelines above are implemented by the LaTeX style file.

3 Figures, Algorithms, and Tables

3.1 Figures

You may want to include figures in the paper to illustrate your approach and results. Such artwork should be centered, legible, and separated from the text. Lines should be dark and at least 0.5 points thick for purposes of reproduction, and text should not appear on a gray background.

Gambar 1: Historical locations and number of accepted papers for International Machine Learning Conferences (ICML 1993 – ICML 2008) and International Workshops on Machine Learning (ML 1988 – ML 1992). At the time this figure was produced, the number of accepted papers for ICML 2008 was unknown and instead estimated.

Label all distinct components of each figure. If the figure takes the form of a graph, then give a name for each axis and include a legend that briefly describes each curve. Do not include a title inside the figure; instead, the caption should serve this function.

Number figures sequentially, placing the figure number and caption *after* the graphics, with at least 0.1 inches of space before the caption and 0.1 inches after it, as in icml-historical. The figure caption should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left. You may float figures to the top or bottom of a column, and you may set wide figures across both columns (use the environment `figure*` in LaTeX). Always place two-column figures at the top or bottom of the page.

Tabel 1: Classification accuracies for naive Bayes and flexible Bayes on various data sets.

Data set	Naive	Flexible	Better?
Breast	95.9± 0.2	96.7± 0.2	√
Cleveland	83.3± 0.6	80.0± 0.6	×
Glass2	61.9± 1.4	83.8± 0.7	√
Credit	74.8± 0.5	78.3± 0.6	
Horse	73.3± 0.9	69.7± 1.0	×
Meta	67.1± 0.6	76.5± 0.5	√
Pima	75.1± 0.6	73.9± 0.5	
Vehicle	44.9± 0.6	61.5± 0.4	√

3.2 Algorithms

If you are using LaTeX, please use the "algorithm" and "algorithmic" environments to format pseudocode. These require the corresponding stylefiles, algorithm.sty and algorithmic.sty, which are supplied with this package. alg:example shows an example.

[tb] Bubble Sort **Input:** data x_i, size m Initialize $noChange = true$. $i = 1$ **to** $m-1$ $x_i > x_{i+1}$ Swap x_i and x_{i+1} $noChange = false$ $noChange$ is $true$

3.3 Tables

You may also want to include tables that summarize material. Like figures, these should be centered, legible, and numbered consecutively. However, place the title *above* the table with at least 0.1 inches of space before the title and the same after it, as in sample-table. The table title should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left.

Tables contain textual material, whereas figures contain graphical material. Specify the contents of each row and column in the table's topmost row. Again, you may float tables to a column's top or bottom, and set wide tables

across both columns. Place two-column tables at the top or bottom of the page.

4 Theorems and such

The preferred way is to number definitions, propositions, lemmas, etc. consecutively, within sections, as shown below.

Definition 4.1 *A function $f : X \to Y$ is injective if for any $x, y \in X$ different, $f(x) \neq f(y)$.*

Using def:inj we immediate get the following result:

Proposition 4.2 *If f is injective mapping a set X to another set Y, the cardinality of Y is at least as large as that of X*

Left as an exercise to the reader. lem:usefullemma stated next will prove to be useful.

Lemma 4.3 *For any $f : X \to Y$ and $g : Y \to Z$ injective functions, $f \circ g$ is injective.*

Theorem 4.4 *If $f : X \to Y$ is bijective, the cardinality of X and Y are the same.*

An easy corollary of thm:bigtheorem is the following:

Corollary 4.5 *If $f : X \to Y$ is bijective, the cardinality of X is at least as large as that of Y.*

Assumption 4.6 *The set X is finite.*

Remark 4.7 *According to some, it is only the finite case (cf. ass:xfinite) that is interesting.*

5 Citations and References

We are going to use the `plainnat` format for references. Citations within the text should include the authors' last names and year. If the authors' names are included in the sentence, place only the year in parentheses, for example when referencing Arthur Samuel's pioneering work lee2023talkin. Otherwise place the entire reference in parentheses with the authors and year separated by a comma, e.g. (for one piece) [**?**]. Please put some effort into making references complete, presentable, and consistent, e.g. use the actual current name of authors. If using bibtex, please protect capital letters of names and abbreviations in titles, for example, use {B}ayesian or {L}ipschitz in your .bib file.

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Authors are kindly asked to make their submissions as accessible as possible for everyone including people with disabilities and sensory or neurological differences. Tips of how to achieve this and what to pay attention to will be provided on the conference website `http://icml.cc/`.

Software and Data

If a paper is accepted, we strongly encourage the publication of software and data with the camera-ready version of the paper whenever appropriate. This can be done by including a URL in the camera-ready copy.

Acknowledgements

Optional acknowledgments go here, as an unnumbered section after the main paper but before the references.

Impact Statement

Authors are should consider including an **optional** statement of the potential broader impact of their work, including its ethical aspects and future societal consequences. This statement should be in an unnumbered and also at the end of the paper, before the references.

A You *can* have an appendix here.